

UST-SREDNEKAN, Russian Federation

WMO: 257050

Lat: 62.4437N

Lon: 152.3211E

Elev: 260

StdP: 98.24

Time Zone: 11.00 (E11)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -46.8	(c) -45.4	(d) -49.5	(e) 0.0	(f) -46.2	(g) -47.9	(h) 0.0	(i) -45.2	(j) 4.1	(k) -23.0	(l) 3.4	(m) -26.9	(n) 0.7	(o) 220	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.1	17.5	26.7	16.5	24.5	15.6	18.7	26.3	17.5	24.6	16.5	22.9	1.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	11.7	21.8	14.7	10.8	20.2	13.7	10.1	19.1	54.4	26.4	50.5	24.9	47.3	22.7	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.2	(b) 3.4	(c) 3.0	(e) -48.0	(f) 33.0	(g) 1.1	(h) 1.6	(i) -48.8	(j) 34.2	(k) -49.4	(l) 35.1	(m) -50.0	(n) 36.0	(o) -50.8	(p) 37.2		
(a) 4.2	(b) 3.4	(c) 3.0	(e) -48.0	(f) 21.0	(g) 1.1	(h) 2.0	(i) -48.8	(j) 22.4	(k) -49.4	(l) 23.6	(m) -50.0	(n) 24.8	(o) -50.8	(p) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.8	-34.2	-31.9	-21.8	-7.7	4.9	13.5	16.8	13.0	5.3	-9.1	-23.4	-32.2		
	DBStd	19.48	7.67	6.97	7.34	7.00	4.81	4.06	4.04	4.18	4.71	5.89	6.68	7.13		
	HDD10.0	7321	1370	1174	985	532	170	14	3	18	153	592	1002	1308		
	HDD18.3	9937	1628	1408	1243	782	417	151	78	172	391	850	1252	1566		
	CDD10.0	468	0	0	0	0	11	120	213	112	12	0	0	0		
	CDD18.3	41	0	0	0	0	0	6	29	6	0	0	0	0		
	CDH23.3	723	0	0	0	0	2	152	462	106	1	0	0	0		
	CDH26.7	207	0	0	0	0	0	32	153	22	0	0	0	0		
(14) Wind	WSAvg	1.3	0.9	1.0	1.2	1.6	1.8	1.6	1.4	1.3	1.4	1.3	1.1	1.0		
(15) Precipitation	PrecAvg	433	33	25	19	16	21	37	70	58	53	40	36	28		
	PrecMax	639	77	47	66	53	64	91	146	156	143	99	77	76		
	PrecMin	289	13	9	4	0	1	7	17	8	5	6	6	7		
	PrecStd	86	17	10	14	12	14	24	38	34	31	23	18	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-17.0	-13.8	-2.3	8.0	23.4	30.0	33.6	29.4	21.4	5.7	-7.3	-12.3		
		MCWB	-17.2	-13.9	-5.1	3.9	13.5	16.9	20.4	18.2	12.7	1.7	-8.6	-12.6		
	2%	DB	-19.1	-17.7	-4.6	5.7	19.5	27.1	30.5	26.2	18.6	2.2	-9.6	-19.0		
		MCWB	-19.4	-18.0	-6.9	1.5	10.5	15.7	18.1	16.7	11.5	0.1	-10.2	-19.2		
	5%	DB	-20.6	-20.2	-7.4	4.3	16.9	24.8	28.1	23.8	15.7	0.1	-11.6	-20.7		
		MCWB	-20.9	-20.5	-9.1	0.5	9.1	14.5	17.4	15.8	9.8	-1.4	-12.1	-20.9		
	10%	DB	-22.7	-22.5	-10.8	2.5	13.7	22.6	25.9	21.3	13.3	-1.4	-13.7	-22.4		
		MCWB	-22.8	-22.8	-11.8	-0.8	7.4	13.5	16.5	14.9	8.9	-2.5	-14.1	-22.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-17.3	-13.9	-4.9	3.8	13.4	18.1	21.2	19.5	13.2	2.2	-8.1	-12.5		
		MCDB	-17.1	-13.7	-2.4	7.6	22.7	27.8	30.8	26.4	19.9	5.4	-7.5	-12.2		
	2%	WB	-19.3	-18.0	-6.9	1.9	11.0	16.8	19.6	17.8	11.9	0.2	-10.2	-19.3		
		MCDB	-19.2	-17.8	-4.7	5.5	18.3	25.2	27.8	24.2	17.0	1.9	-9.6	-19.2		
	5%	WB	-20.9	-20.5	-9.1	0.6	9.3	15.4	18.4	16.6	10.7	-1.3	-12.2	-20.9		
		MCDB	-20.6	-20.2	-7.6	3.8	15.8	23.1	25.9	22.4	15.0	-0.2	-11.7	-20.7		
	10%	WB	-22.9	-22.7	-11.8	-0.5	7.7	14.2	17.3	15.4	9.4	-2.4	-14.1	-22.6		
		MCDB	-22.7	-22.5	-10.7	2.4	13.7	21.1	24.2	20.6	12.7	-1.4	-13.7	-22.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	6.7	11.5	11.9	10.5	12.3	12.1	11.1	8.9	7.0	5.8	5.2		
		MCDBR	4.7	7.4	14.3	10.6	15.7	17.5	17.5	16.4	15.2	8.3	5.3	5.2		
		MCWBR	4.6	7.1	12.3	7.3	8.2	7.6	7.2	7.8	8.3	6.3	5.0	5.1		
	5% WB	MCDBR	4.8	7.2	13.9	10.2	14.6	15.5	15.1	13.8	13.5	7.2	5.4	5.2		
		MCWBR	4.8	7.0	12.0	7.1	7.9	7.5	7.3	7.1	7.7	5.4	5.2	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.191	0.248	0.300	0.369	0.398	0.387	0.451	0.440	0.321	0.276	0.203	0.170		
	taud		2.007	2.117	2.048	1.912	2.075	2.335	2.158	2.135	2.462	2.227	1.994	1.902		
	Ebn at Noon		501	711	788	794	810	835	758	725	789	663	452	306		
	Edh at Noon		33	67	106	152	141	112	130	122	72	59	33	19		
(44) All-Sky Solar Radiation	RadAvg		0.29	1.16	3.03	5.37	5.94	5.53	5.13	3.73	2.29	1.20	0.46	0.12		
	RadStd		0.03	0.09	0.14	0.20	0.41	0.46	0.46	0.40	0.39	0.18	0.05	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page